

Nama: ABIDZAR MAULANA RAMADHAN
NIM: 09020626039
Kelas: H6B.1
Tugas: Pemrograman Dasar

MODUL PEMROGRAMAN DASAR ARRAY JAVA

1. Panjang Array

```
    * @author HYPE AMD
    */
    public class Array {
        //8. panjang array
        public static void main(String[] args) {
            int[] nilai = {80, 75, 90, 85, 70};
            System.out.println(nilai.length); //5
        }
    }
```

Gunakan length, bukan length(), karena length merupakan properti array.

2. Menampilkan Semua Elemen dengan for

```
    *
    * @author HYPE AMD
    */
    //Menampilkan semua Elemen dengan For
    public class SemuaElemenFor {
        public static void main(String[] args) {
            int[] nilai = {80, 75, 90, 85, 70};

            for (int i = 0; i < nilai.length; i++) {
                System.out.println(nilai[i]);
            }
        }
    }
```

3. Input Array dari Keyboard

```
10    */
11    import java.util.Scanner;
12    public class InputArray {
13        public static void main(String[] args) {
14            Scanner input = new Scanner(System.in);
15            int[] nilai = new int[5];
16
17            for (int i = 0; i < nilai.length; i++) {
18                System.out.print("Masukkan nilai ke-" + (i + 1) + ": ");
19                nilai[i] = input.nextInt();
20            }
21
22            System.out.println("\nData nilai:");
23            for (int i = 0; i < nilai.length; i++) {
24                System.out.println(nilai[i]);
25            }
26        }
27    }
```

4. Menghitung Jumlah dan Rata-rata

```

11 public class JumlahRatarata {
12     public static void main(String[] args) {
13
14         int[] nilai = {80, 75, 90, 85, 70};
15
16         int total = 0;
17         for (int i = 0; i < nilai.length; i++) {
18             total += nilai[i];
19         }
20
21         double rataRata = (double) total / nilai.length;
22         System.out.println("Total =" + total);
23         System.out.println("Rata-rata =" + rataRata);
24     }
25 }

```

5. Nilai Maksimum dan Minimum

```

13 public class MaksimumMinimum {
14     public static void main(String[] args) {
15         Scanner input = new Scanner(System.in);
16
17         int[] nilai = {80, 75, 90, 85, 70};
18
19         int max = nilai[0];
20         int min = nilai[0];
21
22         for (int i = 1; i < nilai.length; i++) {
23             if (nilai[i] > max) {
24                 max = nilai[i];
25             }
26             if (nilai[i] < min) {
27                 min = nilai[i];
28             }
29         }
30         System.out.println("Nilai terbesar = " + max);
31         System.out.println("Nilai terkecil = " + min);
32     }
33 }

```

6. Menghitung Jumlah Data Lulus

```

11 public class JumlahDataLulus {
12     public static void main(String[] args) {
13
14         int[] nilai = {80, 65, 90, 70, 85};
15
16         int jumlahLulus = 0;
17
18         for (int i = 0; i < nilai.length; i++) {
19             if (nilai[i] >= 75) {
20                 jumlahLulus++;
21             }
22         }
23
24         System.out.println("Jumlah mahasiswa lulus = " + jumlahLulus);
25     }
26 }

```

7. For each pada Array

```

11 public class ForEach {
12     public static void main(String[] args) {
13
14         int[] nilai = {80, 75, 90, 85, 70};
15
16         for (int n : nilai) {
17             System.out.println(n);
18         }
19     }
20 }
21

```

8. Pencarian Data

```

12
13 public class PencarianData {
14     public static void main(String[] args) {
15         Scanner input = new Scanner(System.in);
16         int[] nilai = {80, 75, 90, 85, 70};
17
18         int cari = 90;
19         boolean ditemukan = false;
20
21         for (int i = 0; i < nilai.length; i++) {
22             if (nilai[i] == cari) {
23                 ditemukan = true;
24                 break;
25             }
26         }
27
28         if (ditemukan) {
29             System.out.println("Data ditemukan");
30         } else {
31             System.out.println("Data tidak ditemukan");
32         }
33     }
34 }
35

```

9. Mencari Posisi Data

```

13 public class MencariPosisiData {
14     public static void main(String[] args) {
15         Scanner input = new Scanner(System.in);
16         int[] nilai = {80, 75, 90, 85, 70};
17
18         int cari = 90;
19         int posisi = -1;
20
21         for (int i = 0; i < nilai.length; i++) {
22             if (nilai[i] == cari) {
23                 posisi = i;
24                 break;
25             }
26         }
27
28         if (posisi != -1) {
29             System.out.println("Data ditemukan pada indeks " + posisi);
30         } else {
31             System.out.println("Data tidak ditemukan");
32         }
33     }
34 }

```

10. Studi Kasus: Nilai Mahasiswa

```

11 import java.util.Scanner;
12
13 public class StudiKasusNilaiMahasiswa {
14     public static void main(String[] args) {
15         Scanner input = new Scanner(System.in);
16         int[] nilai = new int[10];
17         int total = 0;
18
19         for (int i = 0; i < nilai.length; i++) {
20             System.out.print("Nilai mahasiswa ke-" + (i + 1) + ": ");
21             nilai[i] = input.nextInt();
22         }
23
24         int max = nilai[0];
25         int min = nilai[0];
26
27         for (int i = 0; i < nilai.length; i++) {
28             total += nilai[i];
29
30             if (nilai[i] > max) max = nilai[i];
31             if (nilai[i] < min) min = nilai[i];
32         }
33
34         double rataRata = (double) total / nilai.length;
35
36         System.out.println("\n=== HASIL ===");
37         System.out.println("Total = " + total);
38         System.out.println("Rata-rata = " + rataRata);
39         System.out.println("Nilai tertinggi = " + max);
40         System.out.println("Nilai terendah = " + min);
41     }
42 }
43

```

11. Studi Kasus: Penjualan Kantin Kampus

```
10  */
11  public class StudiKasusPenjualanKantinKampus {
12      public static void main(String[] args) {
13          int[] penjualan = {25, 30, 18, 40, 35, 45, 50};
14
15          int total = 0;
16          int tertinggi = penjualan[0];
17          int terendah = penjualan[0];
18          int hariMin30 = 0;
19          for (int i = 0; i < penjualan.length; i++) {
20              total += penjualan[i];
21              if (penjualan[i] > tertinggi) {
22                  tertinggi = penjualan[i];
23              }
24              if (penjualan[i] < terendah) {
25                  terendah = penjualan[i];
26              }
27              if (penjualan[i] >= 30) {
28                  hariMin30++;
29              }
30          }
31          double rataRata = (double) total / penjualan.length;
32          System.out.println("Total Penjualan      : " + total);
33          System.out.println("Rata-rata      : " + rataRata);
34          System.out.println("Penjualan Tertinggi : " + tertinggi);
35          System.out.println("Penjualan Terendah  : " + terendah);
36          System.out.println("Jumlah Hari (>= 30) : " + hariMin30 + " hari");
37      }
38  }
```

12. Studi Kasus: Kehadiran Mahasiswa

```
11  public class StudiKasusKehadiranMahasiswa {
12      public static void main (String[] args) {
13          int[] hadir = {14, 13, 16, 12, 15, 10, 16, 14};
14          int totalPertemuan = 16;
15
16          System.out.println("=== KEHADIRAN MAHASISWA ===");
17          System.out.println("Total Pertemuan: " + totalPertemuan);
18          System.out.println("Batas Minimal : 75% (Minimal hadir "
19              + (int)(totalPertemuan * 0.75) + " kali)");
20          for (int i = 0; i < hadir.length; i++) {
21              double persentase = ((double) hadir[i] / totalPertemuan) * 100;
22              String status;
23              if (persentase >= 75.0) {
24                  status = "Mencapai Target (Lulus)";
25              } else {
26                  status = "Belum Mencapai Target";
27              }
28              System.out.println("Mahasiswa ke-" + (i + 1) +
29                  " | Hadir: " + hadir[i] + "/" + totalPertemuan +
30                  " | " + String.format("%.1f", persentase) + "%" +
31                  " | Status: " + status);
32          }
33      }
34  }
```

13. Studi Kasus: Data Produk UMKM

```

11 public class StudiKasusDataProdukUMKM {
12     public static void main(String[] args) {
13         double[] harga = {15000, 25000, 12000, 30000, 45000};
14         double total = 0;
15         double tertinggi = harga[0];
16         double terendah = harga[0];
17         System.out.println(" LAPORAN HARGA PRODUK UMKM ");
18         System.out.println("No.\tHarga Asli\tDiskon 10%\tHarga Setelah Diskon");
19         for (int i = 0; i < harga.length; i++) {
20             total += harga[i];
21             if (harga[i] > tertinggi) {
22                 tertinggi = harga[i];
23             }
24             if (harga[i] < terendah) {
25                 terendah = harga[i];
26             }
27             double hargaAkhir = harga[i];
28             String infoDiskon = "Tidak ada";
29
30             if (harga[i] >= 30000) {
31                 double diskon = harga[i] * 0.10;
32                 hargaAkhir = harga[i] - diskon;
33                 infoDiskon = "10%";
34             }
35             System.out.println((i + 1) + "\tRp" + (int)harga[i] + "\t\t" + infoDiskon + "\t\tRp" + (int)hargaAkhir);
36         }
37         double rataRata = total / harga.length;
38         System.out.println("Keseluruhan Harga : Rp" + (int)total);
39         System.out.println("Rata-rata Harga : Rp" + (int)rataRata);
40         System.out.println("Harga Tertinggi : Rp" + (int)tertinggi);
41         System.out.println("Harga Terendah : Rp" + (int)terendah);
42     }
43 }

```